Digital Chip Design Program

RTL Design

Almas Alshubrumi

Atheer Al-harbi

Mjd AL-harbi

Maha Omar

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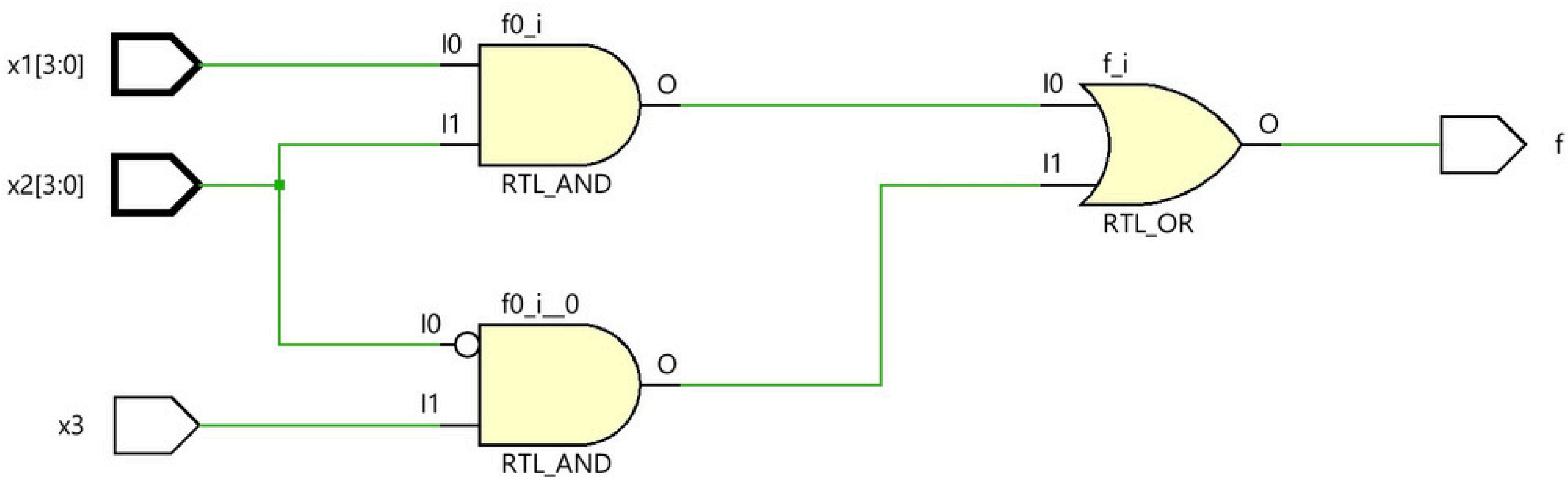
Data Flow

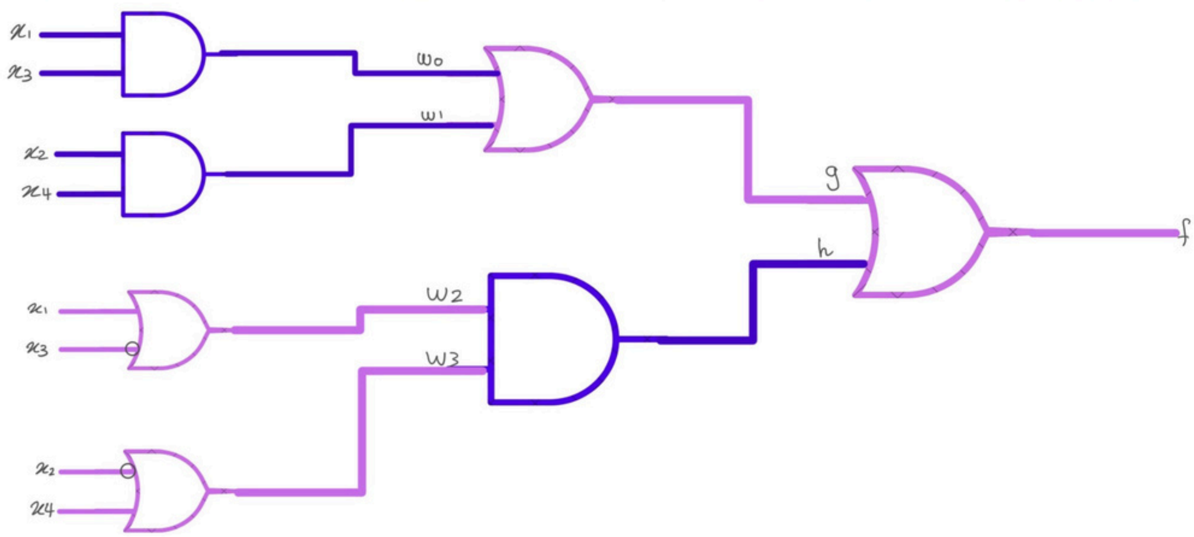
Task #2.1

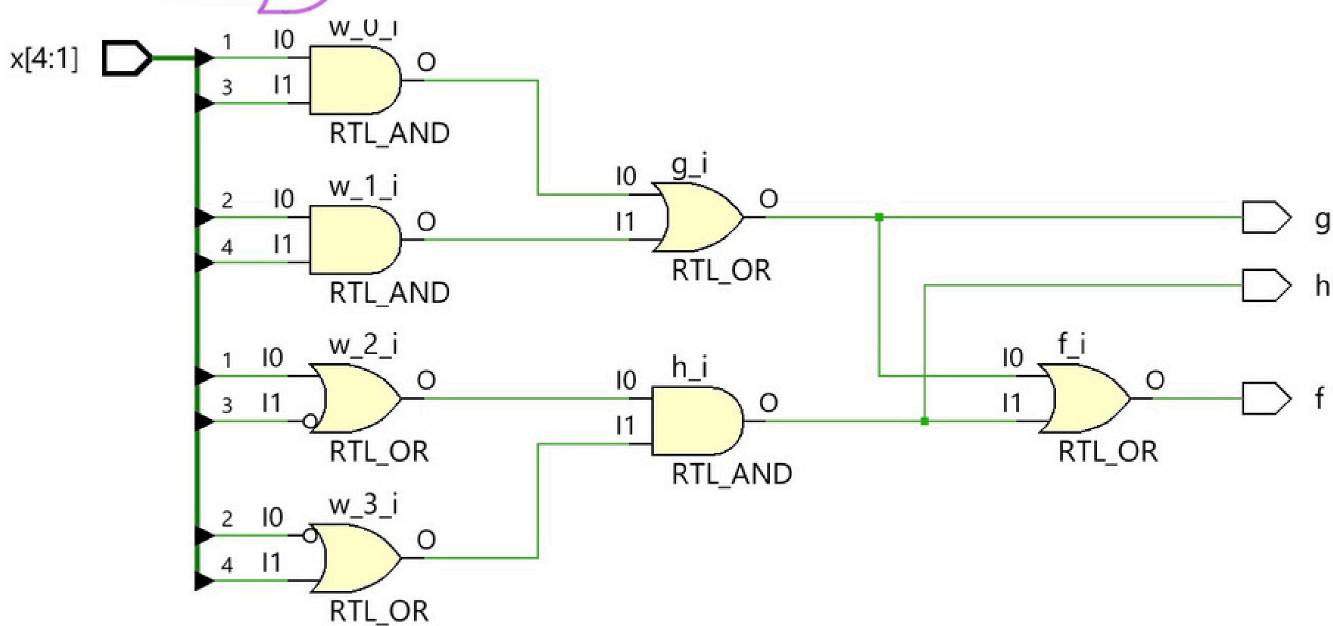
```
module task2
# (parameter n=4)
input logic [n-1:0] x1,
input logic[n-1:0] x2,
input logic x3,
output logicf
) ;

assign f=((x1 & x2) | (~x2&X3)) ;
endmodule
```



Task #2.2

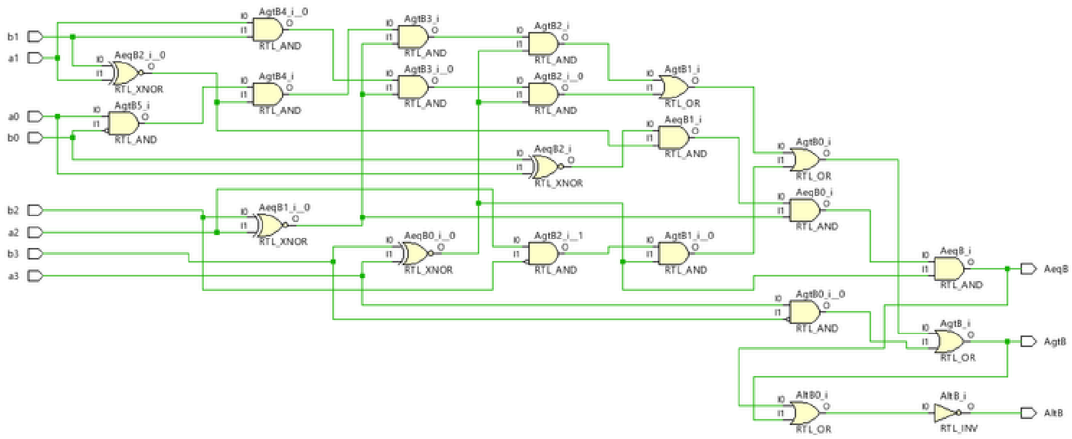
```
module comb_circuit(f, g, h, x);
input [4:1] x;
output f, g, h;
wire [3:0] w;
assign w[0] = x[1] & x[3];
assign w[1] = x[2] & x[4];
assign w[2] = x[1] | ~x[3];
assign w[3] = ~x[2] | x[4];
assign g = w[0] | w[1];
assign h = w[2] & w[3];
assign f = g | h;
endmodule
```

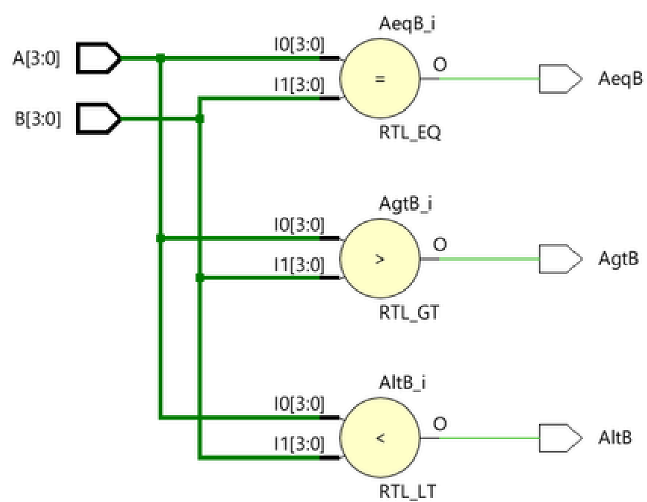


Task #2.3

```
module task23(
input logic a0,
input logic b0,
input logic a1,
input logic b1,
input logic a2,
input logic b2,
input logic a3,
input logic b3,
output logic AeqB,
output logic AltB,
output logic AgtB);
assign AeqB = (b0 ~^a0 )&(b1~^a1)&(b2~^a2)&(b3~^a3) ;
assign AgtB = (( a0)& (~b0) &(b1~^a1)&(b2~^a2)&
(b3~^a3))|((a1)&(b1)&(b2~^a2)&(b3~^a3) )|((a2)&(~b2 )&
(b3~^a3))|((a3)&(~b3));
assign AltB = ~(AeqB | AgtB) ;
endmodule
```




```
module task2_3
# (parameter n=4) (
input logic  [n-1:0] A  ,
input logic  [n-1:0] B,
output logic AeqB,
output logic AltB,
output logic AgtB
    );
    assign AeqB= (A==B);
     assign AltB= (A<B);
      assign AgtB= (A>B);
endmodule
```



Work distribution

Section	Almas	Atheer	Mjd	Maha
Report	✔	✔	✔	✔
Task2.1	✔	✔	✔	✔
Task2.2	✔	✔	✔	✔
Task2.3	✔	✔	✔	✔